



OctoLearn Intelligence Report

Generated By	OctoLearn AutoML	Date	February 22, 2026 at 15:49
Task Type	Regression	Dataset Size	500 rows x 8 features
Risk Score	15/100 (Low Risk)	Best Model	Lightgbm

This report was auto-generated by OctoLearn's intelligent pipeline. It narrates the complete journey of your data -- from raw input through cleaning, profiling, model training, and actionable recommendations.



ACTIONABLE RECOMMENDATIONS

Based on the comprehensive analysis, OctoLearn has generated the following prioritized recommendations to further improve your data and models:

ACTION PLAN FOR PRODUCTION

Strategic Business Insights

- Key Lever: MedInc — Significant statistical weight. Prioritize operational focus on this variable.
- Key Lever: AveOccup — Significant statistical weight. Prioritize operational focus on this variable.
- Key Lever: HouseAge — Driving factor. Age-based segmentation is critical for targeting.

MEDIUM PRIORITY

[MEDIUM] Dataset contains only numeric features. Consider polynomial features or interaction terms to capture non-linear patterns.

[MEDIUM] With 8 numeric features, consider: polynomial features, interactions between important features, and feature scaling/normalization for distance-based models.

INFORMATIONAL PRIORITY

[INFO] No missing values detected in original data. Data quality is excellent.

[INFO] Regression detected. Start with: Linear Regression (baseline), Random Forest Regressor (robust), or XGBoost (best performance). Use RMSE or MAE as evaluation metric.

THE DATA STORY

Your dataset arrived with 500 records and 8 features, setting up a regression challenge. OctoLearn's 6-phase intelligent pipeline examined every corner of this data to extract maximum insight.

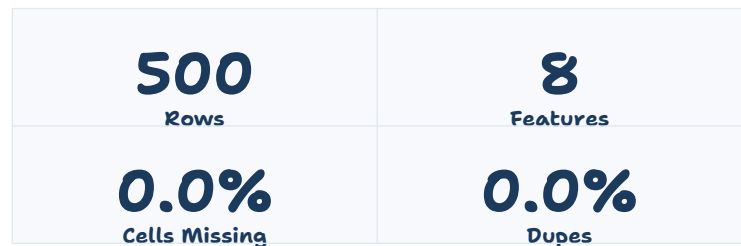
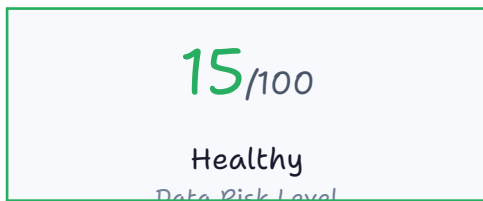
The features break down into 8 numeric and 0 categorical columns. Of the 8 total features, 0 contained missing values that required intelligent imputation strategies.



After deep analysis, OctoLearn assigned a Data Quality Risk Score of 15/100 (Low Risk). This places your dataset in the green zone -- minimal risk factors were detected. Your data is well-structured and ready for modeling.

OctoLearn trained and evaluated 7 models with hyperparameter optimization. After rigorous benchmarking, lightgbm emerged as the champion with a score of 0.5883. See the Model Arena section for full comparison.

DATA HEALTH DASHBOARD



Data Quality Score: 85.0/100



Risk Factors Identified

Factor	Description	Severity
Imbalance	Severe imbalance ratio: 0.032	High

DATA TRANSFORMATION JOURNEY

OctoLearn's intelligent cleaning pipeline automatically transforms your raw data into a clean, model-ready dataset. Here is the before-and-after comparison:

Metric	Before (Raw)	After (Clean)	Change
Rows	500	500	+0
Features	8	8	+0
Numeric Columns	8	8	+0



Categorical Columns	0	0	+0
Cols with Missing Values	0	0	+0

What OctoLearn Did

□ Standardized features using standard scaling

Distribution Comparison: Raw vs Cleaned

The histograms below show how OctoLearn's cleaning pipeline changed the distribution of your top numeric features. Navy = raw data, Green = cleaned data.



Feature Distributions: Before vs After Cleaning



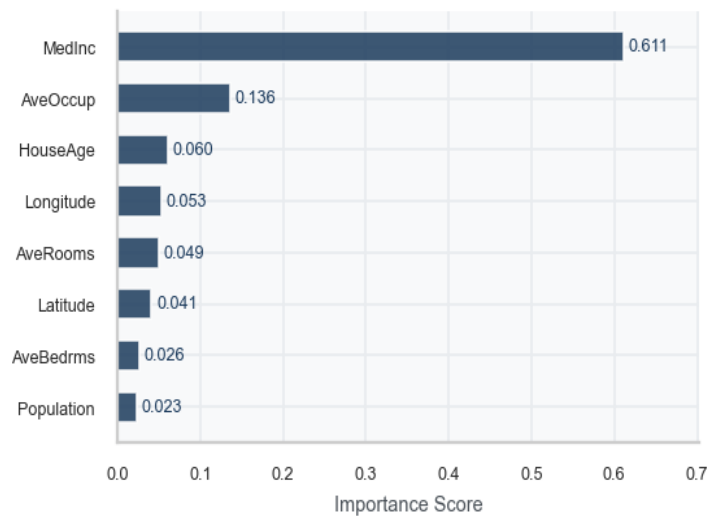
FEATURE INTELLIGENCE

OctoLearn calculated feature importance scores to identify which variables have the strongest predictive power. Higher scores indicate greater influence on the model's



decisions.

Rank	Feature	Score
1	MedInc	0.6110
2	AveOccup	0.1356
3	HouseAge	0.0603
4	Longitude	0.0529
5	AveRooms	0.0492
6	Latitude	0.0413
7	AveBedrms	0.0265
8	Population	0.0231



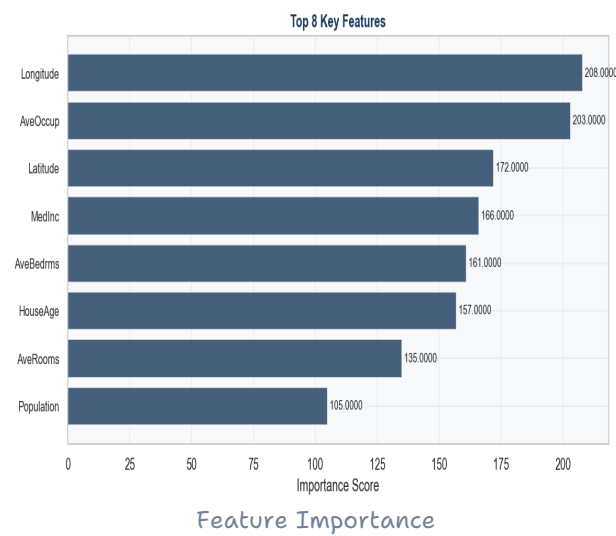
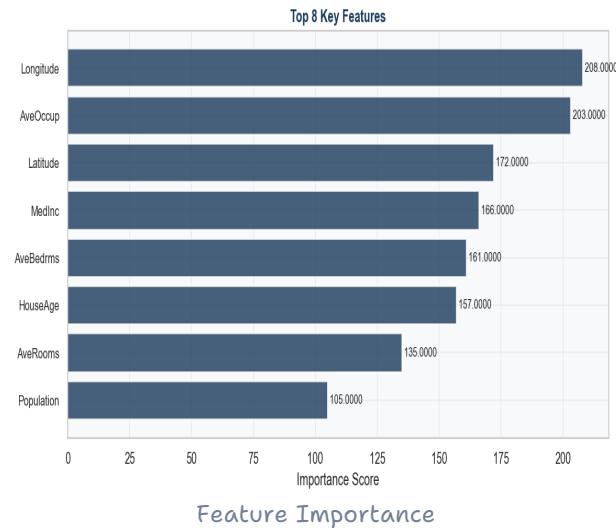
Feature importance scores (higher = more predictive)

MODEL ARENA

Best Model Deep Dive

Detailed performance analysis of the champion model.





OctoLearn trained 7 models with automated hyperparameter tuning. Each model was evaluated on a held-out test set to ensure unbiased comparison. The champion is Lightgbm with a primary score of 0.5883.

CHAMPION
Lightgbm

#	Model	Score	RMSE	MAE	R ²	MAPE
1	lightgbm	0.5883	0.5883	0.4310	0.7526	25.2997
2	stacking_ensemble	0.5965	0.5965	0.3964	0.7457	23.3289



3	xgboost	0.6023	0.6023	0.4050	0.7407	24.066 ₁
4	random_forest	0.6028	0.6028	0.4311	0.7403	27.969 ₇
5	gradient_boosting	0.6229	0.6229	0.4178	0.7227	24.394 ₇
6	linear_regression	0.6591	0.6591	0.4710	0.6895	30.836 ₄
7	svr	1.1952	1.1952	0.9313	-0.0211	58.768 ₂

Models are ranked by primary optimization score. The #1 model (highlighted) was selected as the best performer.

ADVANCED ANALYSIS

Outlier Detection

Detection Method	Outliers Found
Total Outlier Features	5
Total Outlier Rows	106
Outlier Percentage	21.2
Severity	high
Features With Outliers	['MedInc', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup']

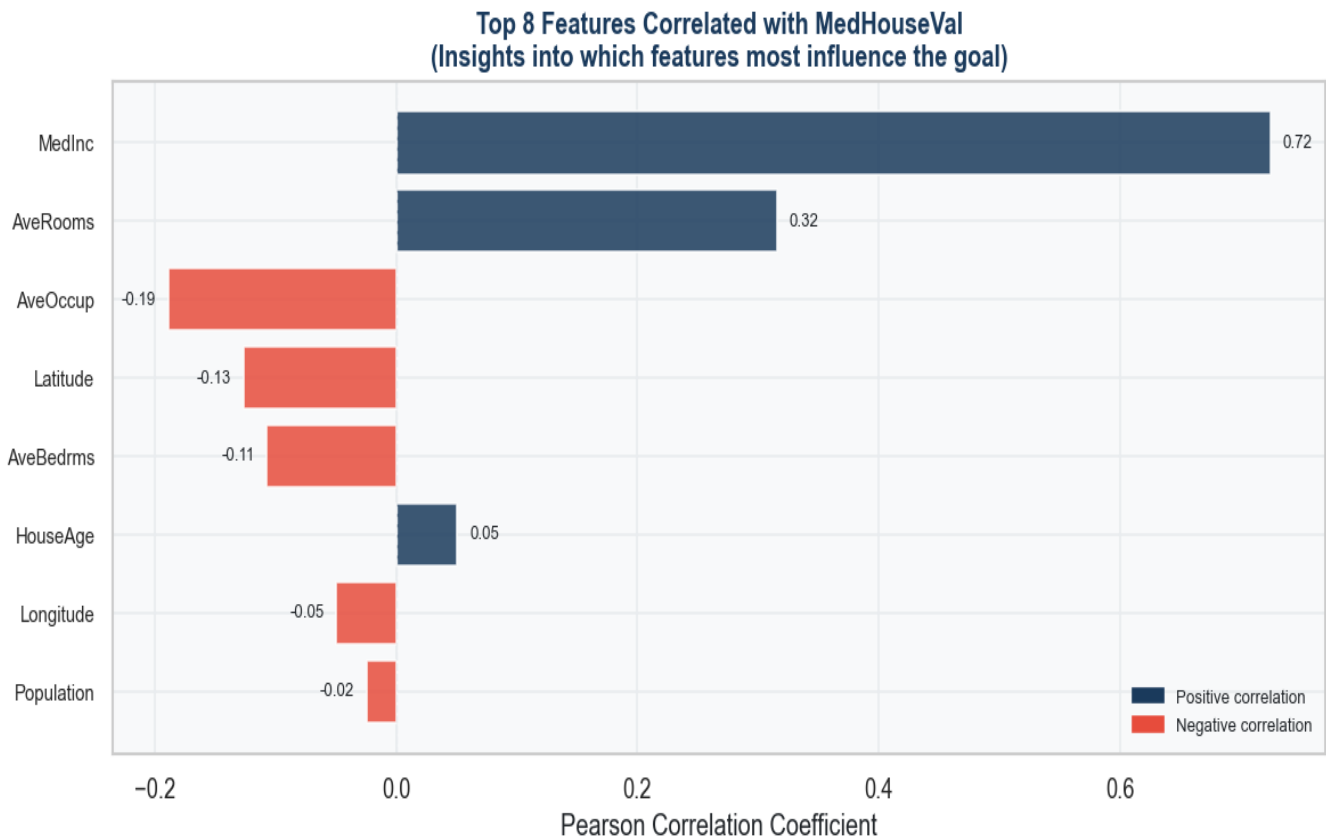
VISUAL INSIGHTS

The following visualizations provide deeper insight into feature distributions and correlations within your cleaned dataset.

Goal Correlation Analysis

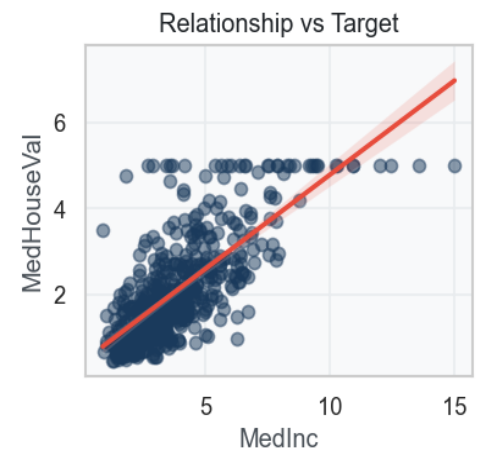
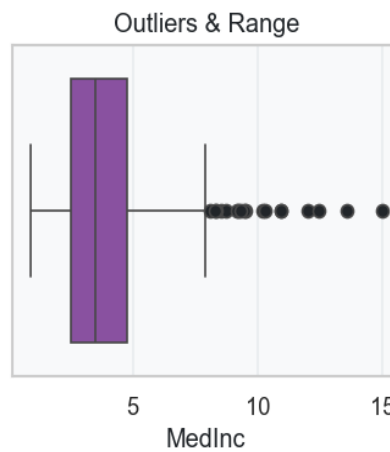
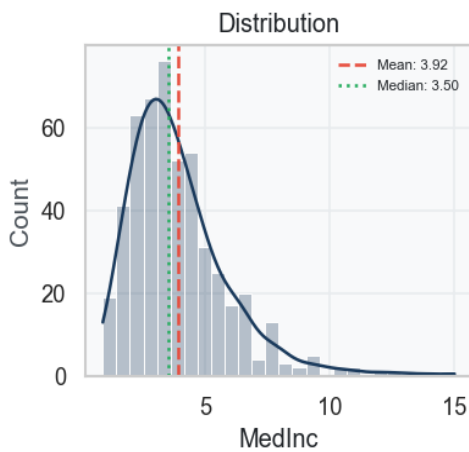
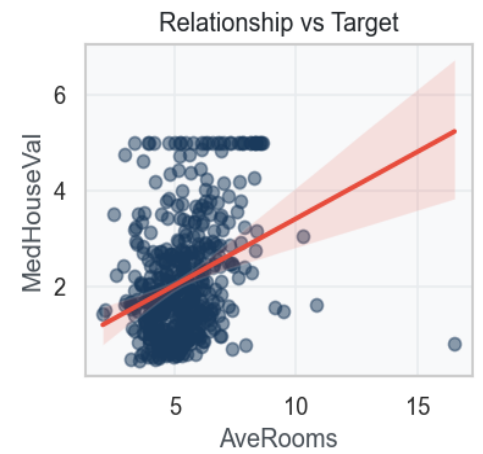
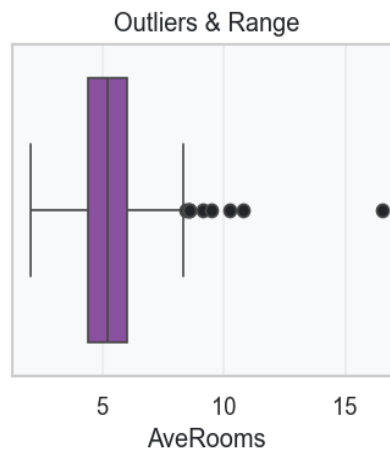
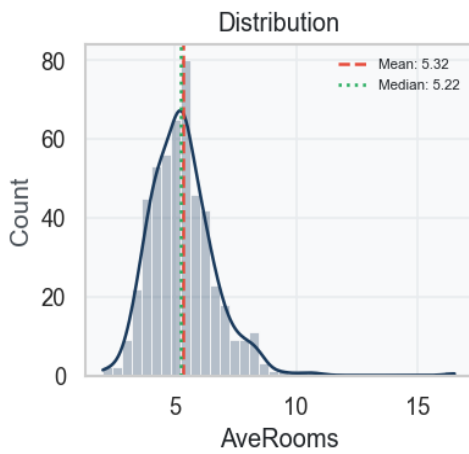
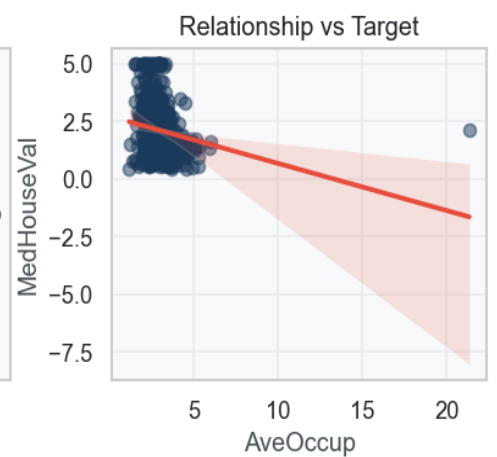
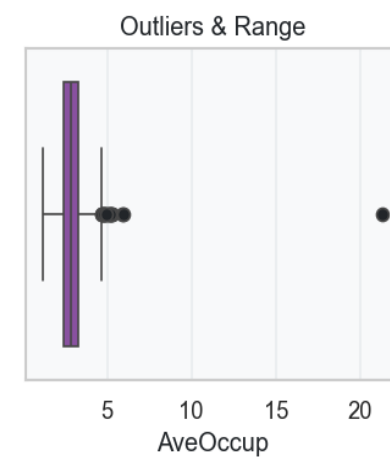
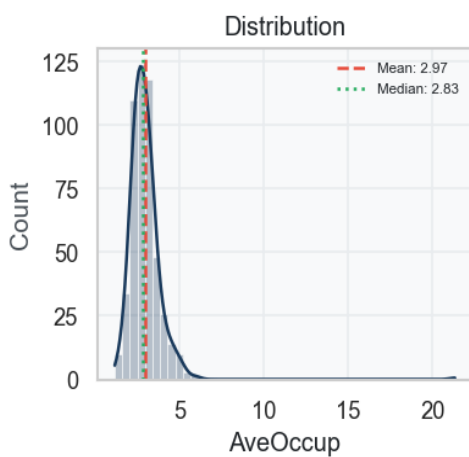


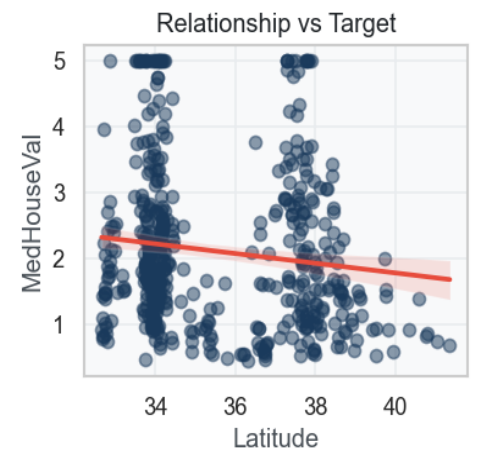
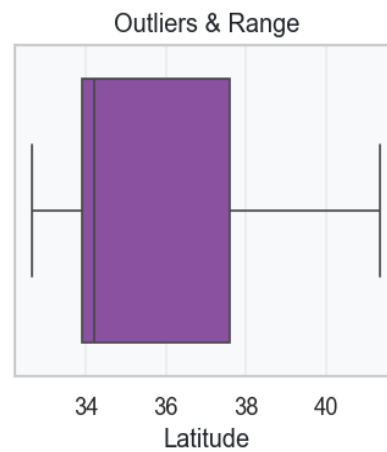
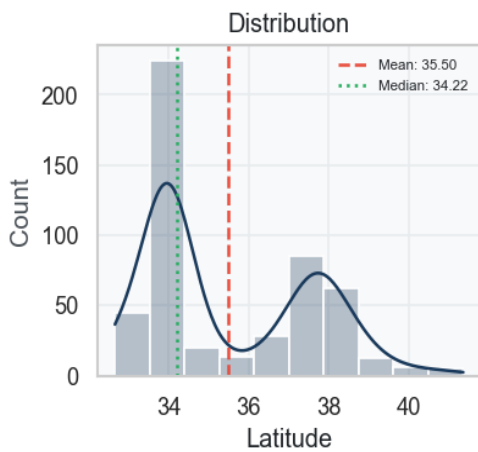
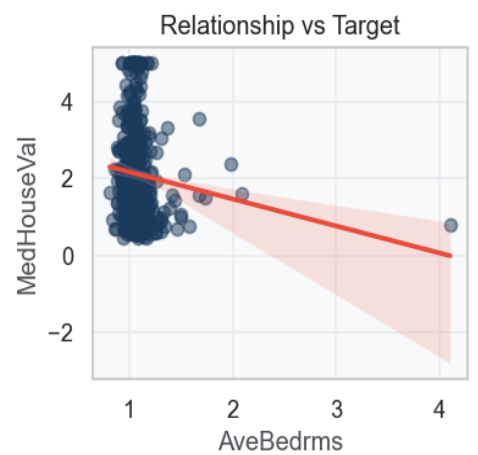
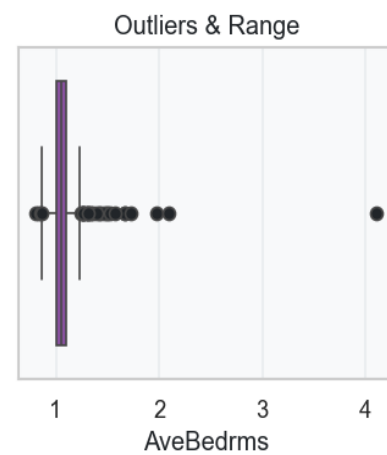
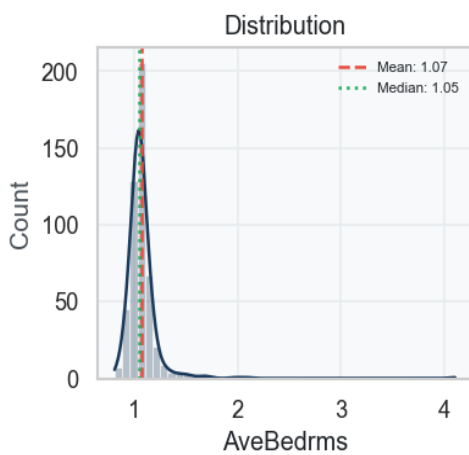
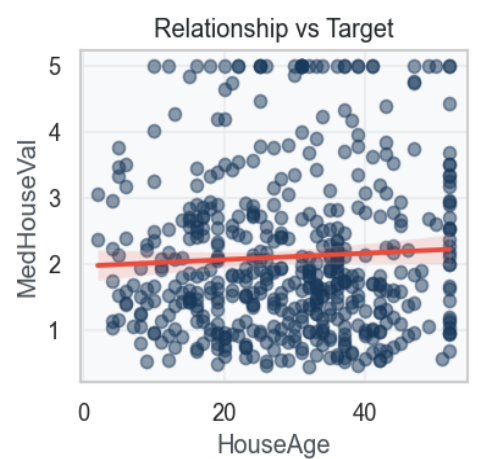
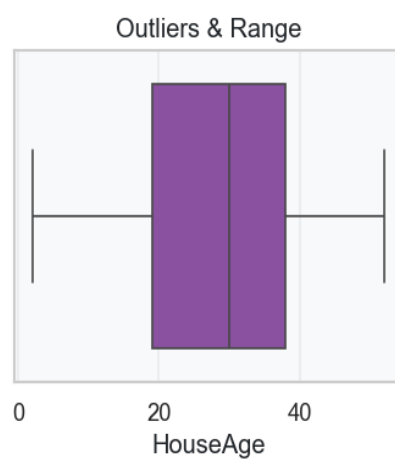
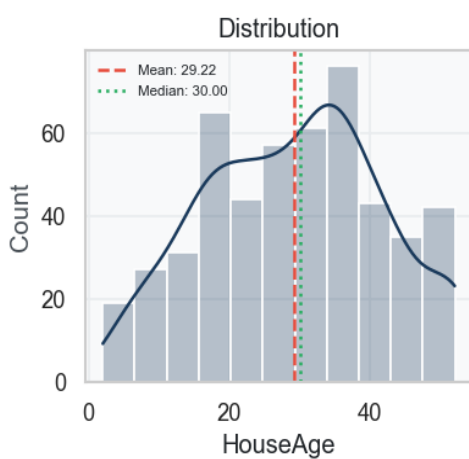
Analysis shows that MedInc (strong positively influence, $r=0.72$), AveRooms (moderate positively influence, $r=0.32$), AveOccup (moderate negatively influence, $r=-0.19$) are the primary predictors for MedHouseVal. Changes in these features are most likely to impact your results. The chart below identifies which of your 8 numeric features most strongly correlate with MedHouseVal. This reveals the key drivers behind your data's goal.



Ranked correlation chart: blue bars = positive correlation, red bars = negative correlation. Higher absolute values indicate features that have more linear influence on the target goal.

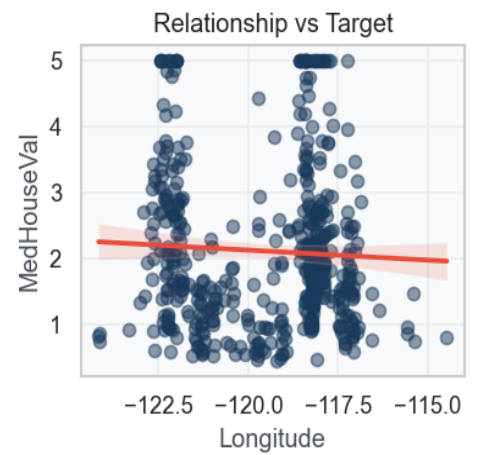
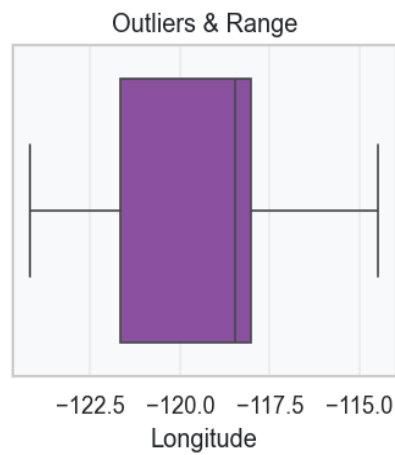
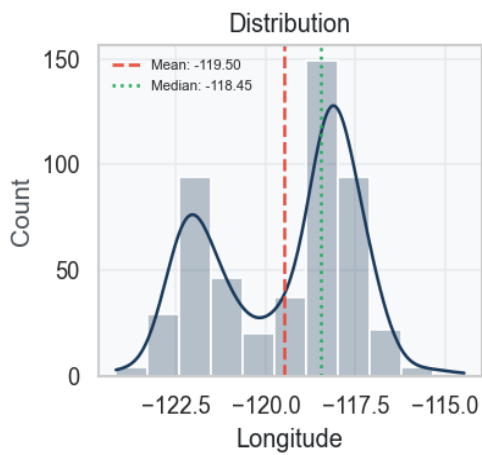
Feature Distributions

**Feature Analysis: MedInc****Feature Analysis: AveRooms****Feature Analysis: AveOccup**

**Feature Analysis: Latitude****Feature Analysis: AveBedrms****Feature Analysis: HouseAge**



Feature Analysis: Longitude



Feature Analysis: Population

